

AESORT Rollout Readiness

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Missing Features and Pilot Risks | Codebase review | September 30, 2026

Scope: current Laravel routes, controllers, models, migrations, and admin/client views. No separate rollout-plan document was present in the workspace; findings are compared against the rollout requirements shared in chat.

Executive Summary

The project has Laravel account, site, device, and subscription CRUD plus a basic KPI dashboard. It is not yet an evidence-backed energy monitoring and savings verification platform. Meter readings, approved baselines, savings evidence, and several operational governance workflows are absent.

Do not present the current sample meter readings or sample report rows as live customer data. Validate Stripe payment server-side before granting a subscription.

Highest-Priority Gaps

- P0 - Real energy data: the client meter screen displays fixed sample values (230 V, 5.2, 1.5 kW, 12.5 kWh, 12 devices, 8 active, 85% load, \$1,240 bill). ClientController::meters
- passes only the user; there is no Meter model or meters/readings table.
- P0 - Payment integrity: Stripe checkout trusts posted amount, plan ID, and site ID. The success route records status=succeeded from session data without retrieving/verifying the Stripe Checkout Session. Checkout is one-time payment while the public plans say per month.
- P0 - Admin password security: current password is not requested or checked. Form sends confirm_password while Laravel confirmed validation expects new_password_confirmation.
- P1 - Device integrity: device create/update accepts status, last_active, and installed_at from the browser; update mass-assigns the full request. Telemetry freshness can be manually overwritten, with no required reason or audit log.
- P1 - Savings governance: no baseline, baseline approval, identified/estimated/implemented/verified savings records, measurement evidence, or verification workflow.

Feature Coverage

- Executive KPI dashboard - PARTIAL. It has aggregate counts and charts, but no verified energy KPIs. Current totals are lifetime counts; trend covers six months; cards say Current month. Card period can therefore be misread as the metric calculation period.
- KPI timestamps - PARTIAL. Latest updated_at/created_at from source tables is shown, not the timestamp of the latest energy ingestion or data pipeline refresh.
- Customer Energy Value - MISSING. No energy usage, cost, carbon, savings, or verified value model.
- Portfolio Benchmarking - MISSING/PARTIAL. Existing site type/area fields are not a

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- versioned benchmark library or normalized performance comparison.
- Analytics & Data - PARTIAL. Registration trends and stored device-status grouping exist;
- actual meter analytics, completeness rules, alerts, and data lineage do not.
- POC/onboarding funnel and customer success - MISSING. No POC stages, milestone owners,
- activation criteria, or recommendation/action tracking.
- Trust, incidents, integration health - MISSING. No severity/response/closure workflow,
- integration registry, ingestion health, or operational run history.
- Audit history - MISSING. audit-logs view is empty; there is no audit model/table or
- active audit route.
- Customer reports - MISSING. Reports & Insights contains a hard-coded example row. No
- customer report export/version/delivery history is implemented.
- Access roles - MISSING. User accounts rely on numeric role 1/2. No granular AESORT
- roles/permissions or admin-role management. User Management view is sample content.

Records and Data Quality

- Sites - PARTIAL. Current fields include customer, address, city/country, area, type,
- timezone, and status. Portfolio, province/state, heating fuel, operating hours,
- data-quality readiness, and baseline status are absent.
- Devices - PARTIAL. Current fields include site, serial, name/type, firmware,
- last_active, status, and installed_at. Asset/equipment link, source unit, expected
- interval, ingestion provenance, and reliable freshness are absent.
- The live database has no meters, readings, baselines, or admin_audit_logs tables.
- Current Site/Device schemas do not include the proposed rollout context fields.

Commercial and Billing

- Admin plan pricing - RISK. Amount and from/to square-foot bands are directly editable;
- validation does not enforce nonnegative prices or valid bands. There is no independent
- approval, plan version/history, or archive safeguard; plans can be hard-deleted.
- Currency and status - PARTIAL. Stripe hardcodes USD, but subscription records do not
- retain an explicit currency code. Payment outcome and subscription term are represented
- by one status field, so Paid and Active/Expired are not independently modeled.
- Public prices are hard-coded separately in subscription.blade.php and do not derive from
- admin plan records. Admin billing route also renders a view that expects subscriptions
- without passing that variable.

Rollout Sequence

1. Replace demo meter values with authenticated, timestamped readings and source/units; build data ingestion and freshness checks.
2. Add site/asset context and data completeness controls; define baseline period, method, version, and independent approval.

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3. Implement savings states and evidence: identified -> estimated -> implemented -> verified, with owner, dates, assumptions, and audit history.
4. Add reasoned audit logs, onboarding/POC stages, recommendations/actions, critical incidents, and integration health workflows.
5. Secure checkout using server-side approved plan pricing and verified Stripe webhooks/session status; separate payment status, subscription term, renewal, and currency.
6. Add role/permission administration, customer report/export management, versioned benchmark library, and acceptance tests for pilot claims.

Verification Notes

The route list currently exposes CRUD and basic pages only; no workbench, audit-log, customer-export, or pricing-approval routes were found. Only the subscription-plan migration and an empty pending operational-context migration are present.

This is a source-code readiness review, not a penetration test, energy-methodology validation, or production deployment assessment.